FURTHER MECHANICS / WORK, ENERGY AND POWER

# Elastic Potential Energy

Questions by topic 10 questions · 101 marks

EXAM BOARDS Edexcel FM1 9FM0/3C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Major Y421 CIE 9231 Paper 3

**Instructions**

Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question gives a value. Where a question leaves $g$ unspecified and a final answer depends on it, the solutions also give that answer for $g = 10\,\mathrm{m\,s^{-2}}$.

CIE (9231 Paper 3): papers take $g = 10\,\mathrm{m\,s^{-2}}$; use the $g = 10$ final answers where the solutions give them.

## Questions


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|---|---|---|
| Question 1 | *6 marks* | 2 |
| Question 2 | *11 marks* | 3 |
| Question 3 | *6 marks* | 4 |
| Question 4 | *9 marks* | 5 |
| Question 5 | *13 marks* | 6 |
| Question 6 | *11 marks* | 7 |
| Question 7 | *12 marks* | 8 |
| Question 8 | *11 marks* | 9 |
| Question 9 | *11 marks* | 10 |
| Question 10 | *11 marks* | 11 |

**1.** One end of a light elastic string, of natural length $2a$ and modulus of elasticity $2mg$, is attached to a fixed point $O$. The other end of the string is attached to a particle $P$ of mass $m$. The string is vertical with $P$ directly above $O$. The particle $P$ is pulled vertically upwards until the extension of the string is $2a$. The particle $P$ is then released from rest.

**(a)** Find the speed of $P$ when it is at a distance $\frac{3}{2}a$ above $O$. **[4]**

**(b)** Find the initial acceleration of $P$ when it is released from rest. **[2]**

**2.** A light elastic string has natural length 2.5 m and modulus of elasticity 15 N.

One end of the elastic string is attached to a particle of mass 0.75 kg.

The other end of the elastic string is attached to a fixed point $O$.

The particle is released from rest at a point $A$, which is 2.9 m vertically below $O$.

**(a)** Calculate the elastic potential energy stored in the string when the particle is at $A$. **[2]**

**(b)** The point $B$ is 4.0 m vertically below $O$.
Calculate the gravitational potential energy lost by the particle as it moves from $A$ to $B$. **[2]**

**(c)** Find the speed of the particle at $B$. **[4]**

**(d)** The point $C$ is 3.3 m vertically below $O$.
Explain, showing any calculations that you make, why the speed of the particle is increasing the first time that the particle is at $C$. **[3]**

**3.** A particle $P$ of mass $m$ is attached to one end of a light elastic string of natural length $a$ and modulus $2mg$. The other end of the string is attached to a fixed point $O$ on a rough plane inclined at $30°$ to the horizontal.

The string lies along a line of greatest slope, with $O$ below $P$. The particle is initially at rest on the plane with the string at its natural length.

The coefficient of friction between $P$ and the plane is $\frac{1}{\sqrt{3}}$.

The particle is projected up the plane, away from $O$, with speed $\sqrt{\frac{3ga}{2}}$.

Find the greatest extension of the string during the subsequent motion. **[6]**

**4.** The points $A$ and $B$ are at the same horizontal level and are $24a$ apart. The ends of a light elastic string, of natural length $18a$ and modulus of elasticity $\lambda$, are attached to $A$ and $B$. A particle $P$ of mass $m$ is attached to the midpoint of the string. The system is in equilibrium with $P$ at a distance $5a$ below $M$, the midpoint of $AB$.

**(a)** Find $\lambda$ in terms of $m$ and $g$. **[3]**

The particle $P$ is pulled down vertically and released from rest from a position $9a$ below $M$.

**(b)** Find, in terms of $a$ and $g$, the speed of $P$ as it passes through its equilibrium position. **[6]**

**5.** A light elastic string has natural length $3a$ and modulus of elasticity $9mg$.
One end of the string is attached to a fixed point $O$. A particle $P$ of mass $m$ is attached to the other end.
The particle hangs freely in equilibrium at the point $E$, vertically below $O$.

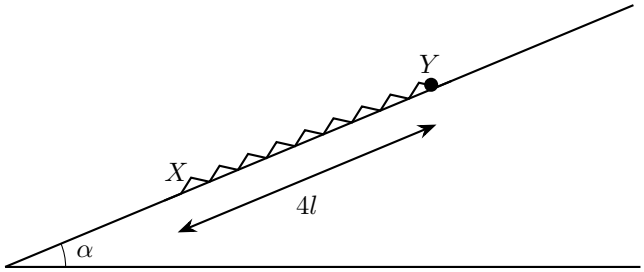
**(a)** Find the length $OE$. **[3]**

The particle is then pulled vertically downwards from $E$ to a point $A$, where $OA = \frac{16a}{3}$, and released from rest.
The resistance to the motion of $P$ is a constant force of magnitude $\frac{1}{4}mg$.

**(b)** Find, in terms of $a$ and $g$, the speed of $P$ as it passes through $E$. **[6]**

The particle is now held at $O$ and released from rest.
The resistance to the motion of $P$ is again a constant force of magnitude $\frac{1}{4}mg$.
The speed of $P$ is greatest when it is at the point $B$.

**(c)** Find the distance $OB$. **[4]**




**6.** A light elastic spring has natural length $2l$ and modulus of elasticity $mg$. One end of the spring is attached to a fixed point $X$ on a rough inclined plane. The other end is attached to a package $P$ of mass $m$.

The plane is inclined to the horizontal at an angle $\alpha$ where

$$\tan\alpha = \frac{5}{12}$$

The package is initially held at the point $Y$ on the plane, where $XY = 4l$. The point $Y$ is higher than $X$ and $XY$ is a line of greatest slope of the plane, as shown. The package is released from rest at $Y$ and moves down the plane.

The coefficient of friction between $P$ and the plane is $\frac{1}{4}$.

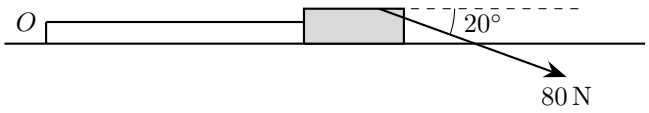
By modelling $P$ as a particle,

**(a)** show that the acceleration of $P$ at the instant when it is released is

$$\frac{15}{13}g$$

down the plane. **[5]**

**(b)** find, in terms of $g$ and $l$, the speed of $P$ at the instant when the spring first regains its natural length. **[6]**




**7.** A block of mass 6 kg rests on a rough horizontal surface and is attached to a fixed point $O$ by a light elastic string. Initially the block is at rest 2.4 m from $O$.

The elastic string has natural length 2 m and modulus of elasticity 80 N.

The coefficient of friction between the block and the surface is 0.25.
A force of magnitude 80 N is applied to the block at an angle of 20° below the horizontal, with its horizontal component directed away from $O$, as shown in the diagram.

The block moves in a straight line directly away from $O$.
In this question, take $g = 9.8\,\text{m}\,\text{s}^{-2}$.

**(a)** Show that the block has moved 0.94 m, correct to 2 significant figures, when the forces acting on it are in equilibrium. **[4]**

**(b)** State one limitation of the model. **[1]**

**(c)** Find the maximum speed of the block. **[7]**

**8.** A spring of natural length $a$ is suspended from a fixed point $A$. A package $P$ of mass $m$ is attached to the lower end of the spring.

The package is initially at the point $B$, vertically below $A$, where $AB = a$.

The package is projected vertically downwards from $B$ with speed $\sqrt{ag}$.

It first comes to instantaneous rest at the point $C$, where $AC = 2a$.

Air resistance is modelled as negligible. By modelling the spring as light and modelling $P$ as a particle,

**(a)** show that the modulus of elasticity of the spring is $3mg$. **[4]**

**(b)** (i) Show that $P$ attains its maximum speed when the extension of the spring is $\frac{1}{3}a$. **[2]**

(ii) Hence find the maximum speed, giving your answer in terms of $a$ and $g$. **[4]**

**(c)** In reality, the spring is not light.
State one way in which this would affect your energy equation in part (b). **[1]**

**9.** A light elastic string with natural length $l$ and modulus of elasticity $kmg$ has one end attached to a fixed point $A$ on a rough inclined plane. The other end of the string is attached to a package of mass $m$. The plane is inclined at an angle $\theta$ to the horizontal, where

$$\tan\theta = \frac{3}{4}$$

The package is initially held at the point $C$ on a line of greatest slope of the plane below $A$, where $AC = 3l$. The package is then projected down the plane with speed $2\sqrt{gl}$ and first comes to rest at the point $B$, where $AB = 5l$.

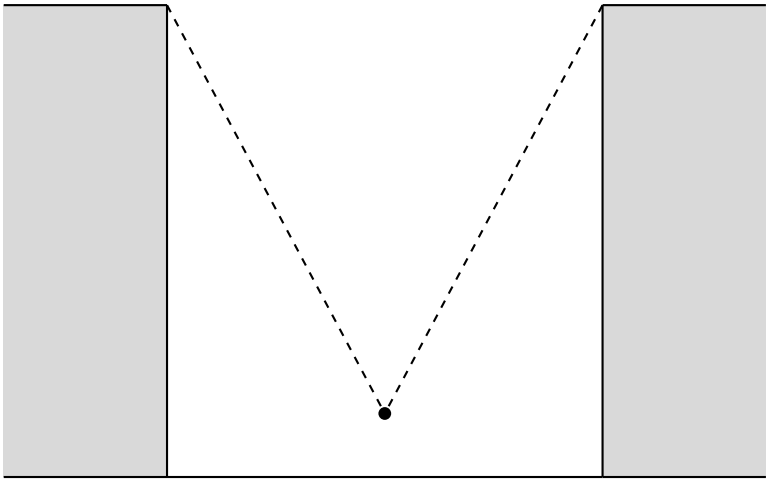
The coefficient of friction between the package and the plane is $\frac{1}{2}$.

By modelling the package as a particle,

**(a)** show that

$$k = \frac{2}{5}$$ **[6]**

**(b)** find the acceleration of the package at the instant it starts to move back up the plane from the point $B$. **[5]**



**10.** A stunt performer is attached to two identical light elastic cords. One end of each cord is fixed to opposite sides of the rim of a vertical shaft, as shown in the diagram above.

The performer has mass 80 kg and is modelled as a particle. Initially the performer is held at rest at the centre of the shaft, 7 m above the floor. The shaft is 52 m deep and 48 m wide. Each elastic cord has natural length 30 m and modulus of elasticity 3150 N.

The performer is released from rest.

Air resistance is modelled as negligible.

**(a)** Show that the speed of the performer when the cords first become slack is $25\,\mathrm{m\,s^{-1}}$ correct to two significant figures. **[6]**

**(b)** Determine whether the performer is moving up or down when the cords become taut again. **[5]**